WORKED SOLUTION

STEP II 1990, Question 1

Pure • Polynomial Inequalities

STEP II 1990, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q1

The key idea

A quartic with no real roots is just a quartic that stays positive, so we look for ways to split it into pieces we *know* are non-negative: perfect squares, or products whose sign is obvious on a particular interval. No single split works everywhere, but different splits work on different intervals, and together they cover the whole real line. The first part hands us the pieces; in the last part we have to find them ourselves.

Solution

The two non-negative expressions

Factorising the quartic and completing the square in the quadratic,

$$x^4 - 2x^3 + x^2 = x^2\left(x^2 - 2x + 1\right) = x^2(x-1)^2,$$
$$x^2 - 8x + 17 = (x-4)^2 + 1.$$

The first is a product of two squares, so it is ≥ 0 for every real x. The second is a square plus 1, so it is at least 1, and in particular non-negative.

The first equation

Write $p(x) = x^4 - 2x^3 + x^2 - 8x + 17$. Notice that p is *not* the sum of the two expressions above (that sum has $2x^2$, not x^2), so we can't simply add them. Instead we split p differently on each interval and show that $p(x) > 0$ throughout.

Ask yourself The question gives two non-negative expressions, but they don't add up to $p(x)$. How can we still use them?

Each of them can be "borrowed" on part of the line: on one interval we keep the quartic $x^2(x-1)^2$ and handle what's left, $17 - 8x$, directly; on another we keep the quadratic $(x-4)^2 + 1$ and handle the leftover $x^4 - 2x^3$ directly.

Case $x \leq 0$. Here $x^4 \geq 0$, $-2x^3 \geq 0$ (since $x^3 \leq 0$), $x^2 \geq 0$ and $-8x \geq 0$. So every term of $p(x)$ is non-negative, and the constant 17 makes the total at least 17. Hence $p(x) > 0$.

Case $0 < x \leq 2$. Write

$$p(x) = x^2(x-1)^2 + (17 - 8x).$$

The first bracket is ≥ 0, and since $x \leq 2$ we have $17 - 8x \geq 17 - 16 = 1$. So $p(x) \geq 1 > 0$.

Case $x > 2$. Write

$$p(x) = x^3(x-2) + \left(x^2 - 8x + 17\right) = x^3(x-2) + (x-4)^2 + 1.$$

Since $x > 2$, both x^3 and $x - 2$ are positive, so $x^3(x-2) > 0$; and $(x-4)^2 + 1 \geq 1$. So $p(x) > 1 > 0$.

The three intervals cover every real number and $p(x) > 0$ on each, so $p(x) = 0$ has no real roots.

The second equation

Let $q(x) = x^4 - x^3 + x^2 - 4x + 4$. We copy the method, now choosing our own split.

Ask yourself Where should we split the line, and which pairs of terms should we group?

We want each group to have an obvious sign on its own interval. The pairing $4 - 4x = 4(1-x)$ is non-negative for $x \leq 1$. For $x \geq 1$, the pairing $x^4 - x^3 = x^3(x-1)$ is non-negative, and what's left, $x^2 - 4x + 4 = (x-2)^2$, is a perfect square. So the natural boundary is $x = 1$.

Case $x \leq 1$. Write

$$q(x) = \left(x^4 - x^3 + x^2\right) + 4(1-x) = x^2\left[\left(x - \tfrac{1}{2}\right)^2 + \tfrac{3}{4}\right] + 4(1-x).$$

The first term is ≥ 0 and is zero only when $x = 0$ (the square bracket is at least $\frac{3}{4}$). The second term is ≥ 0 since $x \leq 1$, and is zero only when $x = 1$. Both terms cannot be zero at once, so $q(x) > 0$.

Case $x \geq 1$. Write

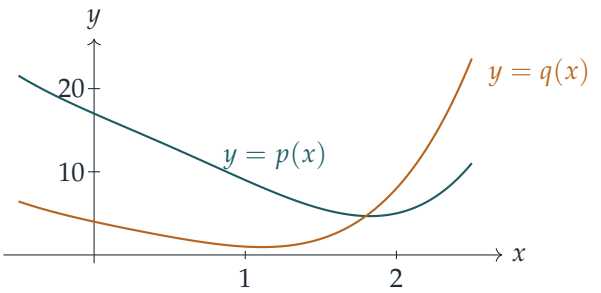
$$q(x) = x^3(x-1) + \left(x^2 - 4x + 4\right) = x^3(x-1) + (x-2)^2.$$

For $x \geq 1$ both x^3 and $x - 1$ are non-negative, so the first term is ≥ 0, and it is zero only at $x = 1$. The second term is a square, zero only at $x = 2$. Again they can't both vanish, so $q(x) > 0$.

So $q(x) > 0$ for all real x, and $q(x) = 0$ has no real roots.

Checks

- Expand each split to make sure it really reproduces p or q. For example, $x^3(x-1)+(x-2)^2 = x^4 - x^3 + x^2 - 4x + 4$.
- A numerical scan of $-5 \le x \le 8$ in steps of 10^{-4} gives a minimum of p of about 4.67 (near $x \approx 1.83$) and a minimum of q of about 0.94 (near $x \approx 1.11$). Both are comfortably positive, as the picture shows.



Common slips

- Adding the two given expressions and claiming the sum is $p(x)$: it isn't, because the x^2 terms don't match.
- Concluding $q(x) > 0$ from "a non-negative term plus a non-negative term". That only gives $q(x) \ge 0$; to rule out a root you must show the two terms can't vanish together.
- Thinking the sign of $-2x^3$ or $-8x$ is always negative. For $x \le 0$ they help rather than hurt.

Beyond STEP

Showing that a polynomial is positive by writing it as a sum of squares is a serious topic. Every non-negative polynomial in one variable is a sum of two squares of real polynomials, but Hilbert showed that this fails for several variables, and this led to his seventeenth problem. Today computers search for such decompositions in optimisation problems. For many more inequality tricks of this kind, Siklos's *Advanced Problems in Mathematics* is a good next step.

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